Supplementary Methods and Results for Main48

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1 Study Workflow and Measurement

Scope recruitment. From an inventory of 3,505 records, we retained 29 earlier scopes and selected 19 additions by source features, public-entry and oracle feasibility, and group-size targets. Recruitment followed inspection of earlier results, and the frame was not fully adjudicated. This is not an independent replication or a probability sample. Selection rules excluded BC/MS, killed counts, timeout rates and correlations. The roster was locked before all 48 scopes were remeasured at ten seeds. Screening counts and dispositions are in the supplementary recruitment audit.

1.1 Oracle Approval and Frozen-Suite Measurement

Suite generation and oracle approval. On Ubuntu with OpenJDK 17.0.20.1, Randoop 4.3.2 [7] generated tests for 60 seconds per scope at seeds 1–10 with `usethreads=false`. Ten seeds fit the compute budget. A five-scope pilot at seeds 101–110 assessed variation in BC, MS and runtime. We compiled suites and replayed them twice in fresh JVMs. One reviewer checked inputs, calls and assertions against the API contract using specifications, source or separate checks. Approved suite source, bytecode, helpers and configuration were then frozen by hash.

Coverage and mutation measurement. The same compiled suite was used for JaCoCo 0.8.11 [3] and PIT 1.30.0 [6] with JUnit 4.13.2 and Hamcrest 1.3. BC is covered/total JaCoCo branches [2]. Raw MS is K/M, where K counts only `KILLED` mutants and M includes all filtered target mutants, including timeouts [5]. We compute MS from XML, not PIT's detected percentage. Hashes were checked before each run. Targets include

Table S1. We assign implementation groups from the eight combinations of source-derived flags.

C	L	R	Label	Scopes
1	0	0	Collection	3
1	1	0	Collection	13
0	1	0	Loop	16
0	0	1	Recursion	7
0	1	1	Recursion	9
1	0	1	Mixed	0
1	1	1	Mixed	0
0	0	0	Outside	0

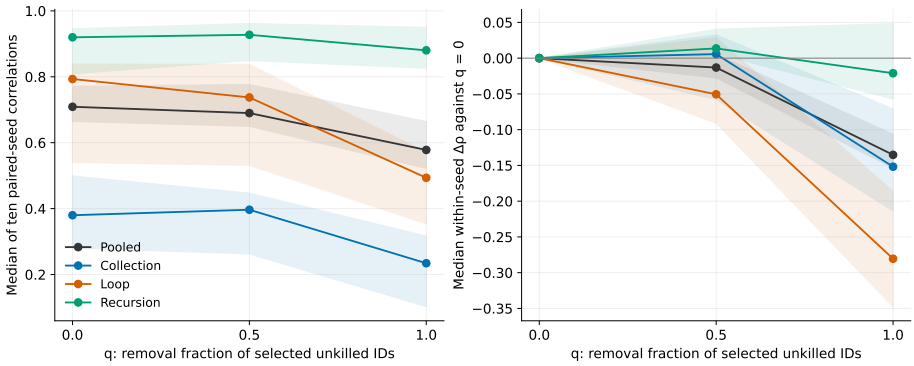


Fig. S1. We show median correlations and paired changes by removal scenario on 48 scopes. Shading spans the ten seeds.

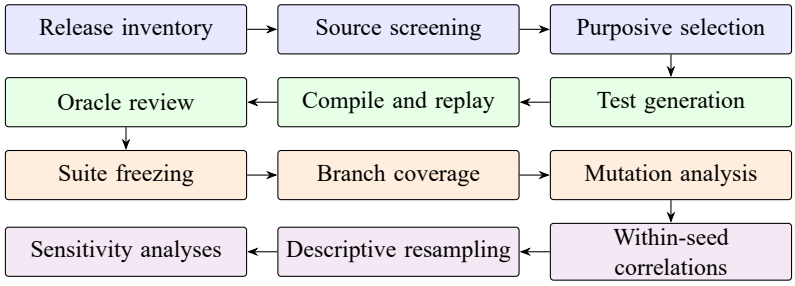


Fig. S2. We recruit scopes, freeze approved suites, measure coverage and mutation outcomes, and analyze within-seed associations.

the declared top-level and listed nested classes, excluding helpers. PIT used one thread, `-Xmx2g`, the locked `DEFAULTS` set (12 resolved mutators), all 33 fixed feature settings, and `timeoutFactor=1.25`, `timeoutConst=3000` ms and a PIT process limit of 1,800 seconds.

Failure handling and retained attempts. No external language-model API is called by the generation or measurement tools. Empty model responses and model-API rate limits are therefore inapplicable. AI writing/code assistance is disclosed separately. Platform API and transfer errors are operational events. Measurements are not automatically re-run, nor suites, seeds or scopes replaced. At most one documented infrastructure retry is permitted with the same configuration and a new, non-overwriting attempt record. Command failures, invalid XML/hashes/universes and technical PIT errors are quarantined, never replaced by zero. Every attempt retains command and configuration records, logs, available raw outputs and failure details. Submission identities are checked before re-submission. Downloads can be retried without remeasurement. No measurement retry was needed. Transport settings are in the supplementary methods notes, and observed event rates in Section 2.4.

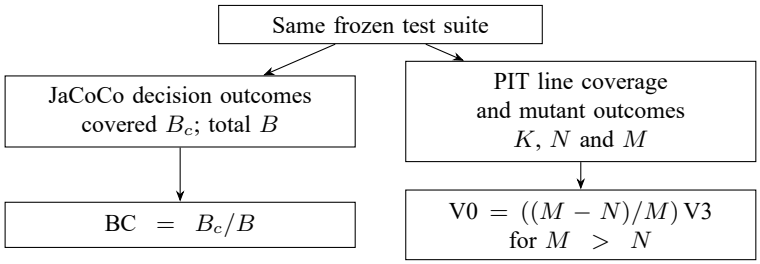


Fig. S3. The metrics share a frozen suite but count different entities. The score identity uses PIT counts, not BC.

Measurement checks. Two XML readers (ElementTree and SAX) compared per-class covered/missed branches, covered-method counts and PIT statuses by full mutant key. The reviewer also performed two script-assisted raw-data checks. All 480 pairs agreed. Frozen records retain targets, seeds, settings and hashes. Both implementations read the same underlying tool outputs.

Branch coverage and mutant-status denominators. Let B_c and B count covered and total JaCoCo branches. For $B > 0$ and $M > N$, the definitions give the exact identity

$$\mathrm{BC} = \frac{B_c}{B}, \qquad \mathrm{V0} = \frac{K}{M} = \underbrace{\frac{M-N}{M}}_{\text{fraction not marked NO_COVERAGE}} \underbrace{\frac{K}{M-N}}_{\text{V3}}. \tag{1}$$

The first factor uses PIT counts, not BC. V3 is not oracle-only because input effects and timeouts remain possible. For $M = N > 0$, $K = 0$ and V0 is zero, but V3 and the product are undefined. JaCoCo decision outcomes and PIT line coverage/bytecode mutants have no one-to-one mapping [2,5] (Fig. S3). The identity implies neither a universal BC relation nor a direction of correlation change.

RQ2b: Selected-key removal scenarios. RQ2b requires an identical full mutant-key universe across seeds within each scope. Eligible keys are `SURVIVED` or `NO_COVERAGE` at least once and never killed or timed out. Keys marked `NON_VIABLE`, `RUN_ERROR` or `MEMORY_ERROR` are also excluded. We select at most ten per scope in lexicographic order of class, method, descriptor, mutator, indexes and blocks: 292 keys in total. No new equivalence classifications are assigned. Links by scope ID and full key to earlier counterexamples identify 22 with prior non-equivalence evidence, including eight in Loop. The same status rule retains them.

For $q \in \{0, 0.5, 1\}$, scope i removes the first $r_i(q) = \lfloor q u_i \rfloor$ of its u_i selected keys and uses $K/(M - r_i(q))$, with the same prefix at every seed. All denominators are positive on the same 48 scopes. Suites, BC, killed counts and mutant universes stay fixed. Subtracting the $q = 0$ coefficient or contrast within seed gives $\Delta\hat{\rho}_{g,s}(q)$ and $\Delta D_{gh,s}(q)$, summarized by ten-seed medians and minimum–maximum ranges. Perturbation summaries count each scope once: eligible/selected keys, scopes with $u_i = 0$ and group distributions of u_i/M_i at $q = 1$. The reported removal percentage is $100u_i/M_i$,

counted once per scope and repeated for both scenario blocks. Changes are calculated within seed before their medians are taken. The ten-key cap makes equal q values unequal denominator fractions. A post hoc check retains the 22 keys with non-equivalence evidence, removes the other 270 at $q = 1$, and recomputes paired changes on all 48 scopes.

Scope of inference. Post hoc evaluation did not establish an adequate inferential procedure for the group comparisons. Given purposive selection and project/package dependence, we could not justify a population weighting and dependence model. We therefore report descriptive estimates and sensitivity analyses for this cohort. The supplementary methods notes document the evaluated procedure and its simulation limitations.

1.2 Resampling and Sensitivity Analyses

We draw scopes with replacement 10,000 times within fixed project-by-group cells, retaining cell sizes and each scope's complete ten-seed vector. Each draw recomputes seed-level correlations and paired differences, then their ten-seed medians. RQ2b uses identical sampled scopes across scenarios to preserve paired changes. Any undefined seed makes that draw statistic undefined. The defined medians' 2.5th and 97.5th percentiles form descriptive ranges. RQ1 checks retained/added subsets, individual projects, leave-one-project-out and omission of each of seven prespecified packages with at least two scopes. We inspect every seed and omit revised PCX-input and Formatters- observer scopes separately and together. Group coefficients require at least three scopes and variation in both metrics.

Post hoc audit of helper use. After viewing the results, we audited source and bytecode hashes for all 480 suites and counted direct helper call sites, called methods and source footprints. Counts include static/instance calls and constructors, not runtime executions. Source and bytecode static-method counts agreed in all pairs. AI-assisted inspection classified input/fixture, observation and observer-setup roles without independent human adjudication. Call sites per test are summarized first by ten-seed median per scope, then across each group. Source footprints include unused shared code. IQR denotes the 25th and 75th percentiles across scopes. Source-unit footprints and distinct called methods count each scope once. The helper supplement contains scope tables, hashes and roles. Supplementary methods notes give counting rules. No Java measurement was repeated.

2 Detailed Results

2.1 RQ2b: Removal of Selected Unkilled Mutants

The rule selected 292 of 1,660 eligible keys, leaving eight scopes unchanged at $q = 1$. Loop had the lowest median removal fraction (Table S2). Fractions describe perturbation size, not its contribution to correlation change. At $q = 0$, correlations and contrasts match main-paper RQ1 table, with zero paired changes. Scenario plots (Fig. S1), eligible-key/no-removal counts and removal-fraction ranges are in the supplement.

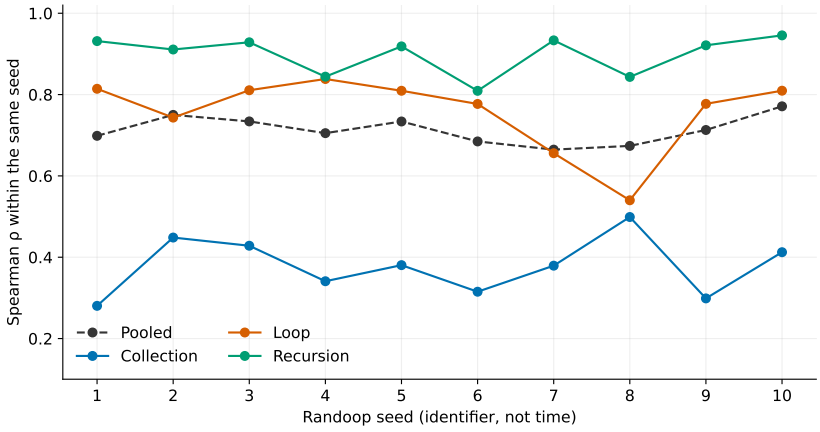


Fig. S4. Each point represents a within-seed BC–MS correlation across 48 pooled scopes or 16 scopes per group.

Table S2. We report correlations and paired changes on the same 48 scopes. Removal fractions describe the full-removal scenario.

q	Group	Selected keys	Median % removed at $q = 1$	Median $\hat{\rho}$	Median $\Delta\hat{\rho}$	$\Delta\hat{\rho}$ range
0.5	Pooled	292	17.5	0.690	−0.013	[−0.027, 0.005]
0.5	Collection	98	21.1	0.397	0.006	[−0.058, 0.032]
0.5	Loop	88	9.9	0.737	−0.051	[−0.091, 0.028]
0.5	Recursion	106	19.3	0.927	0.014	[0.006, 0.040]
1	Pooled	292	17.5	0.578	−0.135	[−0.154, −0.106]
1	Collection	98	21.1	0.234	−0.152	[−0.214, −0.071]
1	Loop	88	9.9	0.494	−0.281	[−0.347, −0.186]
1	Recursion	106	19.3	0.880	−0.021	[−0.057, 0.048]

At $q = 1$, median paired changes were -0.135 pooled and -0.281 for Loop, the largest group decline. Loop-minus-Recursion changed by -0.257 (median ΔD), distinct from its scenario median D of -0.400 (Table S3). In the post hoc $q = 1$ check retaining the 22 keys with prior non-equivalence evidence, the Loop median correlation was 0.624 and its median within-seed change from $q = 0$ was -0.150 (seed range $[-0.230, -0.084]$). The magnitude of the median decline was 46.5% smaller than when all 292 selected keys were removed. One of 10,000 RQ2b draws at $q = 1$ had constant Loop MS at one seed. The Loop coefficient, its two contrasts and their paired changes each used 9,999 defined draws. Point summaries use the original 48 scopes at all ten seeds.

Table S3. We report group contrasts and their paired changes from $q = 0$. Ranges span the ten seeds.

q	Contrast	Median D	Median ΔD	ΔD range
0.5	C minus L	-0.367	0.033	$[-0.049, 0.115]$
0.5	C minus R	-0.540	-0.009	$[-0.082, 0.026]$
0.5	L minus R	-0.188	-0.068	$[-0.106, 0.019]$
1	C minus L	-0.269	0.119	$[-0.027, 0.213]$
1	C minus R	-0.659	-0.128	$[-0.196, -0.014]$
1	L minus R	-0.400	-0.257	$[-0.353, -0.149]$

2.2 Sensitivity to Cohort Composition and Seed

Loop ranked above Recursion on 29 retained scopes, with the reverse on 19 additions (Table S4). Added Collection is N/A: two scopes fall below the minimum of three. Only MOEA's project omission put Loop ahead of Recursion. None of seven package omissions did. Omitting MOEA tree left ten Recursion scopes and narrowed its median to 0.800, versus Loop's 0.793.

Collection ranked highest within JFreeChart, MOEA and POI, whose group cells contain three to six scopes (Table S4). With three untied scopes, Spearman takes only ± 0.5 or ± 1. JabRef's Loop and Recursion cells have fewer than three scopes and are N/A. Collection uses four. As a secondary aggregation check, correlating the separate ten-seed medians of BC and raw MS across the same scopes yielded 0.727 pooled, 0.405 for Collection, 0.815 for Loop and 0.929 for Recursion, preserving the group ordering but not the original within-seed pairing. The primary seed ranges are in main-paper RQ1 table. Omitting Formatters alone raised the Collection median from 0.380 to 0.601. Omitting PCX alone yielded 0.367, and omitting both yielded 0.594. The group order held after each omission.

2.3 Post Hoc Audit of Helper Use

Helper use and call-site rates were highest in Recursion (Table S5). Role counts include any corresponding call across ten seeds and can overlap. Whole-source footprints include shared code not necessarily called by each scope. Use also differed by project. All six POI Recursion scopes and all six MOEA Recursion scopes contained direct helper calls, with group-cell median call sites per test of 19.171 and 1.386, respectively. The helper supplement reports all 48 scopes and the project-by-group breakdown.

2.4 Execution Completeness and Invalid Values

The frozen-main audit covers all 480 planned pairs, each with one attempt and matching XML-reader counts (Table S6). All 2,880 commands (two replays, branch inventory, JaCoCo execution/report and PIT per pair) returned exit code zero. Earlier generation/builder, pilot, adapter and main30 runs are outside these denominators, not assigned

Table S4. We summarize within-seed correlations by cohort subset, project omission and within-project group.

Subset or omission	Scopes	Pooled	Collection	Loop	Recursion
Full cohort	48	0.709	0.380	0.793	0.920
Retained scopes	29	0.623	0.370	0.838	0.786
Newly added scopes	19	0.864	N/A	0.578	0.939
Without JabRef	41	0.861	0.807	0.859	0.902
Without JFreeChart	36	0.658	0.320	0.711	0.964
Without MOEA	33	0.664	0.251	0.820	0.800
Without POI	34	0.655	0.208	0.731	0.944
Without MOEA tree	42	0.677	0.380	0.793	0.800
Within JFreeChart	12	0.892	0.800	0.700	0.500
Within MOEA	15	0.763	1.000	0.846	0.919
Within POI	14	0.946	0.900	0.866	0.829
Within JabRef	7	0.221	−0.769	N/A	N/A

Table S5. We summarize helper use across 16 scopes per group. Call-site rates use ten-seed medians per scope.

Measure	C	L	R
Scopes with direct helper calls	8	9	13
Scopes with input/fixture calls	5	8	13
Scopes with observation calls	4	7	8
Scopes with observer setup	0	0	4
Median whole-source code lines	7	28	132.5
Median declared methods in those units	1	4	15.5
Median distinct directly called methods	0.5	1	2.5
Median helper call sites per test	0.045	1.021	1.775
IQR of call sites per test	[0, 0.453]	[0, 2.592]	[1.024, 11.146]

a zero failure rate. Bootstrap failures preceded worker import because archives were expanded. Repackaging preserved the payload hash and all 48 final kernels completed. The repeated download did not rerun measurement. Its rate is per shard, not attempt, because an early synchronous transfer lacks an attempt counter. `TIMED_OUT` is a retained mutant outcome, not a failed measurement: 87/480 pairs (18.13%) contain timeouts, totaling 690/26,170 mutant–seed outcomes (2.64%).

RQ1/RQ2b use 48 scopes. RQ2a's six exclusions from 48 to 42 reflect undefined V4, not missing executions. The Loop-related undefined draw reported above is separate from measurement errors. Other RQ2b statistics and scenario levels have none. Inferential p-values and population confidence intervals are N/A for every RQ under the descriptive policy in main-paper Methods.

Table S6. We report operational errors, missing data and undefined statistics with their distinct denominators.

Event	Count / denominator	Rate
Pre-worker bootstrap failures / submitted kernels	3 / 51	5.88%
Shards needing a repeated download / final shards	1 / 48	2.08%
Failed or quarantined measurement attempts / attempts	0 / 480	0%
Failed recorded measurement commands / commands	0 / 2,880	0%
Pairs with repeated measurement / planned pairs	0 / 480	0%
Missing final pairs / planned pairs	0 / 480	0%
RQ2a scopes ineligible for common V0–V4 / scopes	6 / 48	12.50%
Undefined RQ1 or RQ2a draws / draws per statistic	0 / 10,000	0%
Undefined RQ2b $q = 1$ Loop-related draws / draws	1 / 10,000	0.01%

3 Additional Validity Details

3.1 Internal Validity

Execution and data integrity. Frozen hashes, two suite replays, two XML readers and two script-assisted raw-data checks address execution integrity. The attempt audit (Section 2.4) separates launch/transfer errors from measurement failures, but cannot detect errors shared by tool outputs. Measurements used several machines, including additions during execution. Fixed settings do not resolve hardware effects on timeouts: no controlled hardware or larger-timeout rerun was performed. Source/binary links rely on release hashes without rebuilding or separately capturing runtime bytecode. One batch lacks its original transfer record. Its hashes were first recorded after download. These hardware and provenance gaps remain unmitigated beyond retained hashes and logs.

Group composition and helper use. Project-by-group counts, subset/omission analyses and input/observer-policy sensitivity expose composition effects. Helper audits describe footprint, calls and roles. Nevertheless, groups share code and features (38 of 48 scopes have the L flag). Without helper ablation or feature-controlled comparisons, causal confounding remains unmitigated. Static call sites exclude runtime use and indirect callbacks. Factories may influence generation without leaving calls in the final suite.

3.2 External Validity

The roster was locked before measurement and recruitment excluded BC/MS values. Subset/omission analyses expose composition sensitivity. However, selection followed earlier results: 48 purposive, overlapping scopes from a partly screened frame of four fixed releases. Prior findings may have influenced recruitment priorities. No independent cohort, other release or generator was evaluated, leaving generalization unmitigated beyond these scopes and the specified Randoop/measurement policies.

3.3 Construct Validity

Metrics and denominator definitions. Explicit KILLED-only scoring, operator/timeout settings, V0–V4 and mutant-status counts define the metrics and distinguish BC from PIT coverage. Common-cohort comparisons separate scoring from eligibility, but excluding six scopes with undefined V4 removes all JabRef timeouts from RQ2a and cannot recover those scores. Equal generation budgets produce different suite sizes. Their influence remains unadjusted and unmitigated [1,4]. No real faults were measured or timeout-budget ablation performed, leaving timeout causes unresolved.

Taxonomy, oracles and mutant equivalence. Two reviewers agreed on 47 of 48 taxonomy group labels before reconciliation (Cohen's $\kappa = 0.969$). Agreement measures consistency within the study cohort. Oracle approval was per seed for Formatters and batched for 470 other suites. Contract checks can miss deviations, and regression assertions can reproduce incorrect behavior. AI-assisted helper-role classification lacks independent human validation. RQ2b uses capped, ordered unknown-mutant scenarios. Retaining 22 keys with prior non-equivalence evidence checks sensitivity, not equivalence. Links use scope ID, full key and source hash. Missing bytecode-hash verification remains unmitigated.

3.4 Conclusion Validity

Within-seed pairing, tied ranks, all ten seeds, explicit NA rules and reported sample sizes/ranges/undefined draws support descriptive reporting. Omission analyses expose composition sensitivity. Nevertheless, project–group cells have at most six scopes, ties/ceilings limit resolution, and singleton resampling cells cannot change membership. Ten seeds were budget-based, not power-based. The five-scope pilot cannot establish full-cohort precision. Repeated seeds reuse scopes and fixed-cell resampling preserves composition, leaving precision and dependence limits unresolved. We therefore report descriptive ranges without strength thresholds. These supply neither population confidence intervals nor independent confirmation.

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